Chromium(charomium and it's extraction, abundance etc).pptx

K kumaravel27122001 Follow

The document provides an overview of the chromium group elements, including their abundance, discovery, extraction processes, and uses. …

1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17

18

CHROMIUM C

More Related Content

Similar to Chromium(charomium and it's extraction, abundance etc).pptx (20)

PPTX

Principles of Metallurgy

pascchemistry

PPTX

Cobalt

Cosnert Rabbani

DOCX

Extraction of metals

MussaOmary3

PPT
Minerals Resources India
Gowri Prabhu

PPTX
Final not yet
XixiViolet

PPTX
Metals
Basecamp Learning Centre

PPT
Occurance n extraction of Al n Fe V1.ppt
LioricLionheart

PPTX
sources and extraction of materials
mustafazariwala

PPTX
Precious elements found on earth and their uses
Rohitsatyaanand

PPTX
hydrogen production and storage details good
Faheem Ahmed

PDF
Metals from Ores: An Introduction-Crimson Publishers
CrimsonPublishersAMMS

PPTX
Uploaded chemistry presentation
Keshav15

PDF
NCERT Chapter- 3 Metals & Non-metals pdf
Vivekanand Anglo Vedic Academy

PPT
Unit 2 periodic table in coursework).ppt
RashmiSanghi

PPTX
Mohun ujoodha metals and their ores form 3

PiLNAfrica

PPTX

Mohun ujoodha metals and their ores form 3

PiLNAfrica

PPT

Unit 2 B Notes Metal Uses Redox 08 09

davedelg1

PPT

Ch22 130105202705-phpapp02

Cleophas Rwemera

PPTX

Occurance of Metals

Dr. Pranabjyoti Das

PPT

Ch24

Universidade Federal de Alfenas

Recently uploaded (20)

PPTX

Artificial Intelligence in Gastroentrology: Advancements and Future Presprec...

AyanHossain

PPT

DRUGS USED IN THERAPY OF SHOCK, Shock Therapy, Treatment or management of shock

Rajshri Ghogare

PPTX

How to Track Skills & Contracts Using Odoo 18 Employee

Celine George

PPTX

Digital Professionalism and Interpersonal Competence

rutvikgediya1

PPTX

YSPH VMOC Special Report - Measles Outbreak Southwest US 7-20-2025.pptx

Yale School of Public Health - The Virtual Medical Operations Center (VMOC)

Chromium(charomium and it's extraction, abundance etc).pptx

1. **CHROMIUM GROUP**
2. **GROUP 6-CHROMIUM GROUP ELEMENTS** – CHROMIUM (^{2} CR[AR]3D 4S^1 ^{4} ^{5} MOLYBDENUM (^{2}MO)- [^{4} KR]4D 5S^1 ^{5} TUNGSTEN (W)- [XE]4F^1 5D 6S^2 ^{74} ^{4} ^{4}
3. **• Chromium is** the twenty – first most abundant element by weight in the earth's crust. • Molybdenum and tungsten are quite rare. ABUNDANCE ELEMENT PPM RELATIVE ABUNDANCE Cr 122 21 Mo 12 56 W 12 56 Abundance of the elements in the earth's crust by weight
4. **DISCOVERY OF CHROMIUM Chromium** was discovered by louis nicolas vauquelin (french chemist) in 1798. He was intrigued by a bright red mineral that had been discovered in a Siberian goldmine in 1766. it was referred to as Siberian red lead and known as crocoite(PbCrO4)
5. **• The element** chromium is named after the Greek word 'chroma' which means colour. • The only commercially important ore of Cr is chromite FeCr O $_2$ 4 • Chromite has a slight lustre and looks like pitch, with a brownish cast to the colour It may be slightly magnetic
6. **EXTRACTION OF CHROMIUM Chromite FeCr O** $_2$ $_4$ NaOH(l) air oxidation Na CrO $_2$ $_4$ O leach Aq.Na CrO $_2$ $_4$ evaporate Solid Na CrO $_2$ $_4$ C, heat reduction Cr O $_2$ $_3$ Si or Al ,heat Cr
7. **Extraction of chromium**
8. **USES Chromium is produced** in two forms ferrochrome and pure Cr metal, For Ferrochrome is an alloy containing Fe, Cr and C FeCr O + C Fe + 2Cr + 4CO $_2$ $_4$ It is used to make many Ferrous alloys, including stainless steel and hard Chromium steel. Ferrochrome
10. **MOLYBDENUM It was discovered** by Carl Wilhelm Scheele in 1778 It was 1st isolated by Swedish chemist Peter Jacob Hjelm in 1781 The name molybdenum comes from Greek word molybdos which mean "lead"
11. **OCCURANCE AND EXTRACTION Molybdenum** occurs as the mineral molybdenite MoS_2 MoS is roasted in air, converting it to MoO . $_2$ $_3$ MoO may be heated with Fe and Al to give $_3$ ferromolybdenum, which is then added to steel. 1. Pure Mo is obtained by dissolving MoO in dilute NH $_3$ 4OH and precipitating ammonium molybdate, dimolybdate or paramolybdate. This is reduced with hydrogen to give the metal Mo
12. **USES Almost 90% of** Mo is used to make cutting steel or stainless steel. Mo metal is used as a catalyst in the petrochemical industry.
14. **TUNGSTEN It was discovered** by "juan and faustro elhuyar" in spain at 1783 The symbol of the element is "W" The w stands for wolfram Tungsten is a rare, naturally occurring silvery-gray metal
15. **OCCURANCE AND EXTRACTION Tungsten** occurs as tungstates, the most common being wolframite FeWO4.MnWO4 and scheelite CaWO4 Different processes are used to extract W from wolframite and scheelite. Wolframite is fused with Na CO $_2$ 3, forming sodium tungstate, which is leached out and acidified to give 'tungstic acid' (the hydrated oxide). Scheelite is acidified with HCl when 'tungstic acid' is precipitated and other materials dissolve. 'Tungstic acid' is then heated to give the anhydrous oxide, which is reduced to give the metal by heating with hydrogen at 850 °C.
16. **USES • Both Mo** and W are alloyed with steel, giving very hard alloys Which are used to make cutting steel. • This is used to make machine tools. • Cutting steel retains its cutting edge even when the metal becomes red hot
18. **Thank you**